# Proposal Evaluation Form

**EUROPEAN COMMISSION**

Horizon Europe Framework Programme (HORIZON)

## Evaluation Summary Report

**Call:** HORIZON-EIC-2025-ACCELERATOR-02

**Type of action:** HORIZON-EIC-ACC

**Proposal number:** 101297706

**Proposal acronym:** AirTransformer

**Duration (months):** 24

**Proposal title:** AirTransformer: Advanced-Porous-Materials-Based Sorption Cooling for Data Centers

**Activity:** EIC

| N. | Proposer name | Country | Total eligible costs | % | Grant Requested | % |
|---|---|---|---|---|---|---|
| 1 | Savery Tech | FR | 2,500,000 | 100.00% | 2,500,000 | 100.00% |
| | Total: | | 2,500,000 | | 2,500,000 | |

**Abstract:**

Europe's data-centre fleet burns 32 TWh of electricity a year—enough to power Denmark—simply to move heat outdoors, and it evaporates billions of litres of potable water, a rising liability as droughts intensify. Traditional options force trade-offs: compressor chillers falter as HFC quotas shrink, while evaporative towers swap power for water many regions can't spare.

AirTransformer breaks that triangle with a new class of crystalline sorbents, whose pore size and chemistry can be tuned molecule-bymolecule.

In a single pass, warm return air from the server hall meets a sorbent bed that captures 5-10 times more moisture per kilogram than silica gels at the same low temperatures. The air leaves cooler and drier; the bed regenerates with the servers' waste-heat and yields distilled water on site. The cycle is compressor-free, PFAS-free, and uses up to 50 % less water than the best adiabatic designs.

Scaling this advanced material is central to the project and fits the EIC Advanced-Materials Challenge:

• Pilot to FOAK line – advance from validated prototypes to a 10 MW production line with continuous novel sorbent synthesis, solvent recovery, and automated bed-packing.

• European value chain – metal precursors, synthesis, assembly, and integration all anchored in EU re-industrialisation regions, insulating against overseas supply shocks.

• Jobs & skills – programme creates 120 direct roles in advanced-material processing and equipment engineering, plus hundreds more in installation and service.

Deployed across just 150 MW of cooling capacity, AirTransformer could save 18 TWh of electricity, 4 million tonnes of $CO_2$, and 5 billion litres of water every year—a direct contribution to the Green Deal, Fit-for-55 energy targets, and Europe's upcoming Water-Resilience agenda. By mastering crystalline adsorbents manufacturing at industrial scale, the EU secures both climate gains and a durable competitive edge in the fastest-growing segment of digital infrastructure.

## Evaluation Summary Report

**Evaluation Result**

**Total score: 0.00 (Threshold: 9 )**

**Scoring explanation**

**The threshold to receive a GO for the proposal at each stage of the evaluation is 9. Companies who do not receive a GO during the remote evaluation and (where applicable) consensus meeting will receive an automatic 0 score. Nonetheless, a breakdown per criterion by each of three experts for the remote evaluation is visible in each section of the Evaluation Summary Report (EXAMPLE: an applicant may receive 2 points in Excellence, 3 in Impact and 1 in Level of risk, implementation, and need for Union support yet the overall score would be 0).**

**An applicant must reach the threshold of 3 points in all 3 criteria to receive a GO overall at both the remote evaluation stage and if applicable in the consensus and at the interview.**

**In the remote evaluation, each criterion receives a score per expert: GO = 1 point and NO GO = 0 points. The threshold to receive a GO at the remote evaluation stage is 9 i.e. 3 evaluators awarding a GO (1 point each) in each of the 3 criteria.**

**For proposals meeting the remote evaluation conditions for consensus meeting, the recommendation to go to interview will be: GO= 9 points or NO GO= 0 points.**

**In the interview, each criterion receives one score from the jury panel of experts: GO= 3 points, NO GO= 0 points. The threshold to receive an overall GO for the proposal at the interview stage is 9 i.e. the panel awards a GO (3 points) in each of the 3 criteria.**

## Criterion 1 - Excellence

Score: **2.00** (Threshold: 3 / 3.00 , Weight: - )

**1. Novelty and breakthrough character of the innovation: Is the proposed innovation highly novel, deep tech breakthrough compared to existing solutions?**

---
*The clarity and pertinence of the proposed project's objectives, and the extent to which the proposed work is breakthrough and goes beyond the state of the art are very good.*
*The proposed project offers a completely new approach to data centre cooling with MOF-based adsorption cooling technology that directly converts low-temperature waste heat (50–70 °C) into cooling energy. Here, metal-organic frameworks (MOFs) are used instead of conventional desiccant materials (such as silica gel, zeolite, lithium chloride), enabling regeneration at low temperatures of 50–70 °C. It is possible to say that the use of MOF materials in conjunction with dew point-controlled indirect evaporative cooling and predictive control algorithms is a high-risk, high-gain deep technology innovation with the potential to transform energy efficiency on a global scale, even though it has not yet been implemented on a large scale. Replacing existing compressor-based or evaporative systems, reducing electricity consumption by 60–70% and water consumption by 40–60%, and achieving this without affecting IT cycles (primary cycles), but only through integration on the air side (secondary cycle), clearly differentiates the project from existing alternatives. The project is an innovation that changes the game by converting waste heat in data centre cooling into an energy input, freeing cooling from dependence on electricity and HFCs. It simultaneously addresses three key challenges facing data centres under rising temperatures and stricter environmental regulations: high electricity consumption, water usage, and HFC dependency. With a payback period of 24–36 months, its modular, plug-and-play design, low maintenance requirements, and easy integration offer superior innovation in terms of cost and ease of use. Finally, the system contributes concretely to both combating climate change and strengthening Europe's sustainable and autonomous material supply chain.*

---
*The clarity and pertinence of the proposed project's objectives, and the extent to which the proposed work is breakthrough and goes beyond the state of the art are very good.*
*The problem and the market opportunity has been described in detailed way. Based on detailed description of alternative solutions of competitors and proposed project it is clear to be convinced that it is an innovation and deep techn. breakthrough solution. The competitors analysis and their solution is reliable and detailed. This is positive. AirTransformer is a ready-to-retrofit response precisely at the intersection of AI booming, water scrutiny and FGas phase-down.*
*The state of the art (part 2.3) is well described, as well as how the proposal will build on it, which is good.*

---
*The clarity and pertinence of the proposed project's objectives, and the extent to which the proposed work is breakthrough and goes beyond the state of the art are very good.*

*The proposed technology is highly innovative as it builds refrigerant-free, water-frugal cooling that turns low-grade waste heat (around 50-70 °C) into reliable latent/sensible cooling for energy-intensive facilities, including data centers.*

*The proposed innovation is convincing in the way it differentiates from competitive solutions. This is evidenced by the useful consultation initiated by the applicants in France and across Europe involving in excess of 200 actors in the data-center cooling value chain, who have clearly articulated the pain points of the cooling industry. The uniqueness of the innovation is convincing in the way it leverages on materials (MOF) to unlock low-grade, heat regeneration, combined with an architecture that keeps ΔP and auxiliaries low while recovering water, supported by a digital twin solution.*

**1.1 Is the applicant applying to a 2025 EIC Accelerator Challenge topic?**

*Yes*

**FOR CHALLENGES ONLY: If yes, does the innovation address the specific objectives of the Challenge to which it was submitted?**

---
*It is understood that the goal of the project is to move metal-organic framework (MOF)-based adsorbents, which are advanced materials, from the lab to industrial production and use them in cooling systems that don't use HFCs and save energy. It therefore makes a real difference in the Challenge's main areas of focus, which are material development, scaling, and commercialization along the whole value chain. The developed MOFs work with regeneration at low temperatures (50–70 °C), which gives them a big energy advantage over traditional materials. This helps Europe become more independent in low-carbon advanced material technologies. The project also uses SSbD and LCA methods to set up the first industrial MOF production line (FOAK line) in the EU and make the supply chain more European. In these ways, AirTransformer is fully in line with both the Challenge's technical goals and its policy goals of making production more sustainable, reducing environmental impact, and strengthening the European materials ecosystem.*

---
*The project directly serves the "Advanced Materials for the Supply Chain" Challenge by industrializing an EU-anchored MOF materials platform, proving safe-and-sustainable performance in demanding field conditions, and standing up a resilient European supply chain for a strategic application: digital-infrastructure cooling.*

---
*The AirTransformer innovation is convincing in the way it addresses the specific objectives of the "Advanced Materials for the Supply Chain" Challenge by industrializing an EU-based MOF-based cooling solution, demonstrating safe-and-sustainable performance under highly-demanding field conditions, and establishing a resilient European supply chain for a strategic application: digital-infrastructure cooling (data centers and edge facilities).*

**2. Timing: Is the innovation at the cutting edge of new market, societal or technological trends?**

---
*The effectiveness and pertinence of the timeline to bring the technology to the market is very good.*
*This project suggests using MOF materials as adsorbents in evaporative cooling systems as a radical way to solve the cooling problem in data centres. The number of data centres has been growing quickly in recent years, and they use a lot of energy and water. This means that it is possible to use MOF to dry the air and low-temperature waste heat sources to do regeneration. It can make evaporative cooling systems work much better if the air is dried, especially in humid places. This will save a lot of water and electricity when cooling data centres. Not using HFCs is also a big help to the environment, though. As artificial intelligence and digitalization become more common, more of these data centres are being built. This means that more energy and water are needed. Every day, the cost of energy and water goes up because of rising demand for energy, ongoing political problems, and pollution of the environment. At this point, the proposed project is being presented as a good solution. It is clear and obvious that there is a need for technology, and the timing is seen as right.*
*It is known that silica gel, zeolite, and lithium chloride are some of the materials that are currently used to remove moisture from evaporative cooling systems used in data centres. The literature also states that MOF materials can be used to get rid of moisture. But none of the current solutions have made a commercial product that uses MOF materials with heat sources that are at low temperatures. In this case, the company has a first-mover advantage. If it doesn't get the right support in time to make its products successful, it could lose its first-mover advantage. There are a lot of studies on MOF materials, which is why this is the case.*

---
*The effectiveness and pertinence of the timeline to bring the technology to the market is very good.*
*Air Transformer is a desiccant-assisted, dew-point IEC DOAS (indirect evaporative cooler as dedicated outdoor air-handling system) module installed on the secondary (air) side. MOF sorbents dry the outdoor air at near-ambient temperature. Waste heat (50–70 °C) regenerates the sorbent. The drier air then feeds an indirect evaporative core, producing 16–20 °C supply air with a controlled dew point of 9–11 g/kg. The result is fewer compressor hours, lower fan power, and significantly less make-up water. This is positive*
*This innovative approach is a materials-led cooling platform that turns the physics of adsorption into a practical advantage for data centres), thereby helping them achieve their climate targets.*
*This is convincing. Especially the detailed description of the solution is clear, well written and reliable.*
*This is very good.*

---
*The effectiveness and pertinence of the timeline to bring the technology to the market is very good. The survey with key players in the European Data Center Cooling value-chain is convincing.*

*This is convincingly substantiated by several factors, including the rising rack heat densities and AI-driven workloads that are pushing sensible and, especially, latent loads beyond what airflow or control configurations can absorb. Also, latent control alone still accounts for circa 30–60% of cooling energy, combined with water-scarcity situations in several countries in Europe. In this context, Data Center operators now face a credible challenge to deliver simultaneous reductions in electric load and water use, while eliminating refrigerant risk.*

**3. Technological feasibility: Has the technology been developed in a safe, secure and reliable manner? Has it been adequately assessed, validated or certified? Is there sufficient demonstration that the innovation has completed all aspects of TRL 5 (validation in a relevant environment for the application of the technology) and therefore the innovation is currently at TRL 6?**

---
*The credibility of the pathways taken to test and validate the technology and complete all aspects of TRL 5 is very good.*
*The project is technically sound, and the basic operating principles are reliably shown through laboratory tests at TRL-5 level (integrated cartridge (MOF contactor + dew-point IEC + regeneration loop) tests, data based on experiments that follow ASHRAE-143/EN13141-7 protocols). The project can now be rated as TRL-6, even though independent ISO/IEC-17025 accredited third-party testing is still going on. The proposal outlines a solid plan for development and certification (CE /EMC, REACH/SSbD, M&V protocols) for low-temperature regeneration, ΔP targets, dew-point control, and field integration. It also points out important TRL-related risks, such as the durability of MOF materials (≥10,000 cycles), the ability to scale production (FOAK line), and long-term field performance. It also plans for tests and mitigation activities that will go along with these risks (long-cycle tests, fouling/filtration strategies, spare module architecture). The project's technical reproducibility and fault management approach (FMEA, go/no-go criteria, maintenance/service procedures) is available in draft form. The Legionella/Water Safety Plan and SSbD commitments are good for society and the environment. In short, the proposed technology is based on sound science and has a clear plan for how to get there.*

---
*The credibility of the pathways taken to test and validate the technology and complete all aspects of TRL 5 is very good.*
*The solution has been developed over the last six years, based on researches in the lab. The description of observed outcomes is sufficient.*
*The activities of each TRL levels has been well described (part 3.3).*
*The TRL-related future plans has been also provided, which is positive.*
*The project is technically robust, accurate and reproducible, and able to deal with and inform about possible failures. It is described mainly in risk analysis. Which is positive*
*The project is socially robust and duly consider the context and environment in which they will operate. This is well described, also with relevance to EU politics, which is positive.*

---
*The credibility of the pathways taken to test and validate the technology and complete all aspects of TRL 5 is very good.*

*The technology has been convincingly developed in a safe, secure and reliable controlled psychometric lab across hot-humid and mild-wet rating points. For example, the innovation subsystem was tested with a working outdoor air stream and a recirculating product stream, delivering stable supply dew-points in the 9-11 g/kg range and tempered supply at 16-20 °C. Instrumentation and data reduction are credible in the way they followed ASHRAE 143 (IEC) and EN 13141-7 (air-side performance) with ISO/IEC 17025 calibration traceability.*

*As such, the innovation has been adequately assessed and validated. Formal product certifications in relation to Eurovent DC AHU label and CE LVD/EMC are credibly planned in relation to the grant element of the proposal.*

*There is convincing demonstration that the innovation has completed all aspects of TRL 5, i.e., validation in a relevant environment for the application of the technology. The applicants are currently advancing toward TRL 6. For example, a fully instrumented prototype is undergoing independent validation under controlled psychometric conditions to demonstrate repeatable performance in a relevant operational environment.*

**4. Intellectual Property Strategy: Does the company have the necessary Intellectual Property Rights to ensure adequate protection of the idea? Has a freedom to operate analysis been carried out?**

---
*The soundness of the arguments proving the proposal's intellectual property strategy is average.*
*The proposal generally gives a good structure for an intellectual property strategy, but there are some important areas where things are unclear. The project uses basic MOF formulations and synthesis processes that are licensed by CNRS and only SaveryTech can use them. This protects the core technology. The proposal also says that it includes two international patent applications (in the PCT phase) for a system architecture based on moisture adsorption and low-temperature regeneration principles. It also says that this protection has been extended to "air-handling integration" modules. The fact that the Freedom to Operate (FTO) analysis has only been done at a preliminary screening level, though, could be a problem when entering the market. A thorough review of current patents for comparable MOF-based systems (e.g., BASF, NuMat, Framergy) in the US and Asian markets is essential. The company's intellectual property protection strategy is not very clear. It includes things like confidentiality agreements (NDAs), joint development agreements (JDAs), and limits on sharing data. However, it needs to be clearer about IP ownership, license scope, and how new applications will be added to the commercialization plan. The project appendix does not include the license agreement with CNRS. So, the scope of the agreement is not sufficiently convincing. In conclusion, even though the current IP assets seem to protect the core technology well enough, an FTO analysis that guarantees "all freedoms necessary for successful commercialization", a current patent map, and a strategic IP plan for non-European markets should have been shown.*

---
*The soundness of the arguments proving the proposal's intellectual property strategy is very good. IP strategy is clear and convincing.The company IP portfolio has 3 PCT patents,. It is positive.*
*Savery Tech owns the machine, the cycle, and the controls; their academic partners own the chemistry of the adsorbents.*
*On brand and commercial enablement, they are registering AirTransformer, and Savery Tech as EUIPO/UKIPO trademarks, with copyright protection on firmware and HMI assets and, where appropriate, design registrations on industrial enclosures and cartridges. Which is positive.*
*It is stated the there is plan when their pilots mature over the next 12–24 months, they will file continuations/divisionals to extend the protection perimeter to heating /heat-reuse configurations and industrial latent control which is positive*

---
*The soundness of the arguments proving the proposal's intellectual property strategy is very good.*

*The applicants are convincing in that they have the necessary Intellectual Property Rights to ensure adequate protection of their innovation.The IP landscape of the innovation involves on the materials side reliance on a license owned by a third-part organisation, CNRS, in relation to MOF sorbents optimized for water adsorption with low-temperature regeneration in industrial cooling and dehumidification. This doesn't constitute a risk as the applicants possess an exclusive EU field-of-use licence from CNRS, including sub-licensing rights to qualify manufacturing partners, and a performance-based diligence clause aligned with the applicants ramp plan, ensuring the rights are exercised at pace while giving them freedom to build a European supply chain.*

*A convincing freedom to operate analysis has been carried out. The FTO is conclusive in that based on clear recommendations: (a) the need to clearly distinguishes the two claim spaces (two-MOF HMX vs. MOF for closed-cycle adsorption); (b) isolates the true risk drivers (geometric IEC/HMX claims; D2/D3 chemistry /substrate families), and (c) suggesting practical design around and "safer lane" strategies; the applicants should have presently the freedom to Operate in Europe.*

## Criterion 2 - Impact

Score: **3.00** (Threshold: 3 / 3.00 , Weight: - )

**5. Customer Demand: Is the solution bringing sufficient added value to trigger demand from potential customers?**

---
*The soundness of the arguments proving the proposal's competitiveness and added value is very good.*
*The AirTransformer solution gives data centres and industrial facilities a measurable extra value that will make customers want it by cutting cooling costs and environmental impact by a large amount. Current solutions, like compressor-based chiller systems, evaporative coolers, or absorption cooling units, aren't very energy-efficient or long-lasting because they use a lot of electricity, water, or high-temperature heat for regeneration. AirTransformer, on the other hand, only uses low-temperature waste heat in the 50–70 °C range. This is possible because of MOF-based adsorption technology. This cuts electricity use by 60–70% and water use by 40–60%. It also runs without any HFCs, which means it follows all environmental rules. Non-invasive "no-touch retrofit" modules make integration easy, which makes it great for existing facilities that don't need to go offline. This means low investment risk and a short payback period of 24–36 months for businesses. Its biggest advantages over other technologies are that it is cheap to run, it offers a lot of benefits (like lowering energy use, water use, and HFC emissions), and it follows the rules. In these ways, the project is a great deal for both data centre operators and industries that use a lot of energy. It has a good chance of being accepted by the market because energy costs are going up and there is pressure to be more environmentally friendly. These advantages are sufficiently detailed and convincingly explained by the proposer.*

---
*The soundness of the arguments proving the proposal's competitiveness and added value is very good.*
*The main expected indicators have been qualified and are presented in detailed way and they are credible. The proposal addresses the criterion very well.*
*The expected impacts are credible and justified with regard to the role, past accomplishments and engagement of the partners.*
*The impacts during and after the project are clear, they are in line with the topic and ambitious. The impacts are credible based on the foreseen activities; this is very good.*
*The company started with several customers (including early customers). Which is very positive*
*The innovation is better than the existing/alternative solutions and is described in reliable and sufficient.*

---
*The soundness of the arguments proving the proposal's competitiveness and added value is very good.*

*The proposed solution is credible in its potential to bring sufficient added value to trigger demand from potential customers. It operates at the intersection of advanced HVAC and waste-heat-for-cooling and offers the clear advantage of a retrofit-friendly solution, compared to the competition.*

*The applicants have clearly evidenced the added-value their solution offers. As an example, their solution provides (a) climate and humidity swing costs by millions per year for the same 10 MW*
*IT load, and (b) even in temperate cities, utilities alone are in average €3 M/yr; without factoring in maintenance, chemicals, labor, and refrigerant risk, that increase this cost estimate.*

**6. Market development: Does the innovation have the potential to develop new markets or significantly transform existing ones? Has the potential market for the innovation been adequately quantified, including conditions and growth rates? Is the expected market share acquisition reasonably ambitious and reachable?**

---
*The credibility of the market development plan and targeted innovation area description are very good.*
*The AirTransformer solution could change the way things are done in the data centre cooling market and open up a whole new group of cooling technologies that can use low-temperature waste heat while using less water and electricity. Compressor-based and HFC-based systems are very important to this market right now, and evaporative coolers are also important, but not as much. Current technologies have reached their limits when it comes to being environmentally friendly because they use a lot of electricity and have a big carbon and water footprint. By using low-temperature waste heat directly for cooling, AirTransformer is creating a new value chain in this field. The project aims for a market opportunity of about €9 billion in Europe by 2030, which is the same as 7 GW of new data centre cooling capacity. It also predicts that the total accessible market will reach €25 billion with low-temperature waste heat applications in industries like food, chemicals, glass, and paper. These estimates are in line with the EU's rules on energy efficiency and the current rate of growth (12–15% per year). The project plans to start in the European data centre market with pilot applications and a licensed production line. Then, it will grow globally by working with system integrators. The commercialization strategy is clear: TRL-7 field demonstrations, a FOAK production line, certification (CE, ISO 50001), and partnerships with OEMs. Considering the targeted market areas, current technological trends, and regulatory dynamics, the project appears realistic and achievable, with the potential to both transform the existing cooling market and create a new green cooling segment based on waste heat.*

---
*The credibility of the market development plan and targeted innovation area description are very good.*
*Market development has been presented including marketing and sales plan, competitors analysis, and go-to market plan with development of new markets. It is described in sufficient way and it is very good.*
*Have been conducted a comprehensive study of the market, ensuring a more thorough and detailed assessment, including competitors analysis (part. 4.3) This makes the proposal attractive.*

---
*The credibility of the market development plan and targeted innovation area description are very good.*

*The innovation has the potential to significantly transform the existing infrastructure cooling market, with a focus on data centers. The combination of (a) refrigerant-free intervention, (b) ultra-low-water, (c) operational range of 40–60 °C waste heat, and (d) secondary-side retrofit is convincingly unique in the European data-center landscape.*

*The potential market for the innovation has been adequately quantified, including conditions and growth rates. This is evidenced by the convincing projected global baseline and growth context. As an example, the applicants project the industrial cooling / waste-heat-enabled cooling segment to expand from $20.1B in 2024 to $28.7B in 2029, a total increase of circa $8.6B (+43%), equivalent to circa $1.7B average annual uplift, implying a circa 7.4% CAGR over 2024–2029.*

*The expected market share acquisition is credible and reasonably ambitious and reachable. The applicants projected global SAM is convincingly substantiated and estimated at circa 0.2-0.6 % over 36 months.*

**7. Commercialisation strategy: Is there a convincing and well thought, thorough strategy for commercialisation, including regulatory approvals /compliance needed, time to market/deployment, and business and revenue model? Are the key partners identified and committed?**

---
*The soundness of the arguments proving the proposal's commercialisation strategy is very good.*
*The project mostly meets the expected parts for the market entry of technology-focused deep innovations and gives a good plan for commercialization. The project lays out a three-phase plan for exploitation: (1) Proof of performance and reliability through field demonstrations at TRL-7/8 level, (2) Commissioning of the first FOAK MOF production line in Europe and certification of module assembly processes (CE, EMC, REACH, SSbD, ISO 50001), and (3) Licensing and transition to serial production scale through system integrators and OEM partnerships. This strategy makes a phased plan for entering the market, taking into account compliance with regulations (especially F-gas, Ecodesign, and energy efficiency standards). It also expects a "time-to-market" period of about three years. During the production and field application phases, the company made initial deals and partnership commitments with CNRS, IFPEN, data centre operators, and HVAC system integrators. The business model is based on licensing and revenue sharing instead of direct sales. This model allows for quick growth while lowering the risk of production costs. In general, the commercialization strategy is well thought out, the partnership structure is stable, and regulatory compliance is taken into account.*

---
*The soundness of the arguments proving the proposal's commercialisation strategy is very good.*
*The strategy covers several aspects including marketing approach, customer acquisition proces, sales proces and pricing strategy, dissemination plan.*
*This is very good. Such approach is reliable and suitable for such solutions.*

---
*The soundness of the arguments proving the proposal's commercialisation strategy is very good.*

*The proposed commercialisation strategy is convincing and well thought thorough. For instance, the business revenue model is clearly substantiated and as such convincing. The innovation will be sold as modular, retrofit-friendly equipment for data-center cooling and dedicated outdoor air handling system for industrial /commercial scenarios, complemented by services and software. For instance, the cost of the Hardware is €1 100/kW (pilot series) trending toward €900/kW at ≥150 MW/year.*

*Regulatory approval is clearly identified and will form the focus of the grant element of the project as credibly substantiated in the proposed work plan.*

*The identified value chain comprising key partners to support the commercialisation ambition of the applicants is appropriate. This is evidenced by the secured non-binding letters of intent, that signal the association with key market players.*

**8. Scale up potential: Does the innovation have the potential to scale up the company?**

---
*The soundness of the arguments proving the scale up potential is very good.*
*The AirTransformer project gives SaveryTech a lot of room to grow and can help the company make money in the short term and position itself strategically in the long term. If the project is successful in commercialization, the company could move from its current structure focused on research and development to a business model that can grow based on manufacturing, licensing, and services. The proposal aims to reach an annual turnover of €20–25 million within 5 years and create more than 60 direct jobs. These goals seem reasonable. The proposed funding need—around €10–12 million for setting up the FOAK production line, field demonstrations, and certification processes—is realistic and reasonable given the technical and industrial scale needs. The timeline (three years from TRL-6 to TRL-8, then two years for series production and first license sales) is also appropriate. Overall, the project has the potential to make SaveryTech a leader in advanced material-based sustainable cooling solutions in Europe. It also has the potential to grow both technologically and financially. The scale up potential of the company is very high and a detailed and convincing time and financing plan has been presented to realise this.*

---
*The soundness of the arguments proving the scale up potential is very good.*
*- the scale-up technology possibilities - it is possible to scale-up the project to bigger one.*
*- the financial needs are realistic and well described - detailed WP description and Gantt chart - which is positive.*
*- the business idea is convincing and contains a reasonable timeline - 36 months - to reach TRL 8.*
*- the scale-up potential is described in several aspects: turnover, capacity, licensing, team and management, risk and financial plan which is positive.*

---
*The soundness of the arguments proving the scale up potential is very good.*

*The impact the innovation will have on the company is convincing, and clearly evidenced in terms of growth and scale-up potential.*

*The company has a clear potential to scale-up. This is evidenced by clear projections, during the grant delivery and beyond.*

*The estimated costs and financial needs to develop the idea are realistic and credible. These have been clearly substantiated for both the grant and investment elements.*

*The business idea is convincing and contains a reasonable timeline clearly substantiated in the proposal. This is structured into waves with pertinent Key Performance Indicators.*

**8.1 Scale up potential: Is the applicant applying for Grant Only support?**

*No*

**For grant only support: If yes, can the applicant demonstrate access to the resources needed to commercialise and scale-up the innovation?**

*Not provided*

**9. Broader impact: : Will the innovation, if successfully commercialised achieve positive societal, economic, environmental or climate impacts?**

---
*The soundness and relevance of metrics supporting the innovation's potential societal/economic/environmental/climate impact is very good.*
*If the AirTransformer solution is successfully commercialized, it could have big and measurable effects on the environment, the economy, and society as a whole. Compared to traditional compressor-based systems, this technology uses 60–70% less electricity and 40–60% less water. This directly relieves pressure on carbon emissions and water resources. These effects are directly in line with the goals of the European Green Deal, climate change mitigation, and energy supply security. The project is expected to create high-value-added jobs and production in the advanced materials and sustainable cooling sectors in Europe. It will also strengthen the domestic supply chain and make Europe less reliant on imports. It will also help the data economy grow in a way that is good for the environment by lowering the energy load that comes with cooling. It will also encourage the spread of green facility ideas. The metrics that the application asks for—energy intensity (PUE), water intensity (WUE), $CO_2$ reduction, and number of jobs—are clear, measurable, and convincing.*

---
*The soundness and relevance of metrics supporting the innovation's potential societal/economic/environmental/climate impact is very good.*
*Societal, economic, environmental and/or climate impacts are taken into consideration and properly described.*
*It is worth to mention that wide relevance to EU polices has been considered as important reason to develop the solution. This is very good.*

---
*The soundness and relevance of metrics supporting the innovation's potential societal/economic/environmental/climate impact is very good.*

*The impact is clearly articulated, with a clear methodology involving inputs, activities, outputs, and outcomes, supported by pertinent Key Performance Indicators. This is convincing and substantiated across (a) the environmental and climate dimensions, with clear metrics such as Electricity saved,, $CO_2$e avoided (energy), Peak-load shaved, Heat valorized, Water saved, HFC leakage avoided, and Air pollutants avoided; and (b) Societal & economic impacts, with clear metrics, such as service continuity, opex freed for innovation, as well as yearly job creation over a 5 years period. As such, credible examples are given across social, environmental, economic and climate dimensions, supported by credible quantitative metrics.*

*The proposal presents credible metrics on how the innovation will potentially contribute to the addressed Challenge. As an example, their solution has the potential to cut data-center operators cooling electricity by circa 60-70% and water withdrawal by circa 40-60% by turning low-grade waste heat (50-70 °C) into useful cooling with MOF-based sorption.*

**9.1: Is the applicant applying to a 2025 EIC Accelerator Challenge topic?**

*Yes*

**FOR CHALLENGES ONLY:**
**If yes, please enter your comments for each of the following questions:**
**- Does the innovation have the potential to contribute to the expected outcomes and impacts set out in the Challenge to which the proposal was submitted?**
**- Does the innovation have a clear and measurable positive impact on the EU Single Market (in line with STEP objectives), either by:**
**• Strengthening its competitiveness: Does the innovation address a critical need within the EU internal market, fostering innovation and technological leadership for European companies?**
**• Reducing strategic dependencies: Does the innovation contribute to reducing the EU's reliance on non-EU sources for critical technologies or raw materials?**

**(Note: applicants from Associated Countries will need to demonstrate contributions to the EU internal market or reduction of strategic dependencies)**

---
*The AirTransformer project is a strong new idea that can help directly with the results and effects that the "Acceleration of advanced materials development and upscaling along the value chain" Challenge wants to see. The project meets the main goal of the Challenge, which is to safely, sustainably, and competitively move advanced materials from the lab to the factory. It does this by trying to bring MOF-based advanced materials with low-temperature regeneration capabilities to industrial scale in Europe. The project could help the EU market become more innovative and technologically advanced, which is in line with STEP goals. Developing a European technology-based alternative, particularly in a rapidly growing sector such as data centre cooling, will create a strategic competitive advantage. The project also helps the EU reach its strategic autonomy goals by building a local supply chain and raw material processing infrastructure for making MOFs. This makes the EU less reliant on advanced material suppliers in Asia and the US. AirTransformer could be an example of how to make real, measurable, and systemic changes to both the EU's climate and industrial policies if it is successfully commercialized. It could have an effect on environmental sustainability, energy efficiency, and material security.*

---
*- Does the innovation have the potential to contribute to the expected outcomes and impacts set out in the Challenge to which the proposal was submitted? YES*
*- Does the innovation have a clear and measurable positive impact on the EU Single Market (in line with STEP objectives), either by:*
*• Strengthening its competitiveness: Does the innovation address a critical need within the EU internal market, fostering innovation and technological leadership for European companies? YES*
*• Reducing strategic dependencies: Does the innovation contribute to reducing the EU's reliance on non-EU sources for critical technologies or raw materials? YES*

---
*The innovation has a credible potential to contribute to the expected outcomes and impacts set out in the addressed Challenge. As an example, their solution relies on metal–organic framework (MOF) composites and engineered sorbent architectures that enable low-grade-heat-driven dehumidification / cooling for data centers, evidencing advanced materials at the core.*

*The innovation has a clear, convincing and measurable positive impact on the EU Single Market. This is evidenced by (a) its credible drive to strengthen EU competitiveness in EU-based MOF media and system assembly, reducing imports, creating exportable deep-tech products, and helping European data-center operators meet tightening efficiency and disclosure rules at lower total cost of ownership; (b) addressing a critical need for the data center cooling needs within the EU internal market, fostering MOF-based innovation and technological leadership for the company; and (c) reducing strategic dependencies on non-EU sources for critical raw materials by establishing dual-sourcing for key MOF materials, engaging EU-based fabrication/assembly partners, and commissioning a quality-assured first-of-a-kind (FOAK) MOF production line within the Union, thereby mitigating export-control and logistics shocks.*

## Criterion 3 - Level of risk, implementation, and need for Union support

Score: **2.00** (Threshold: 3 / 3.00 , Weight: - )

**10. Team: Does the team have the capability and motivation to implement the innovation proposal and bring it to the market? Is there a plan to acquire any critical competencies which are currently missing, including adequate gender balance?**

---
*The capacity and role of each team member, the measures in place to ensure motivation and the extent to which the proposed team, as a whole, brings together the necessary knowledge and expertise are very good.*
*It is possible to say that the project team is very good at technical skills, industry experience, and the desire to make money, and they have the skills to bring new ideas to market. The SaveryTech team is made up of a core group of people who are experts in different areas, such as adsorption systems, advanced materials synthesis (especially MOF chemistry), thermal system design, and managing energy use in data centres. Additionally, collaborations with institutions such as CNRS and IFPEN provide access to high technical capacity in R&D and scaling. Roles within the team, such as R&D leader, process engineer, business development manager, regulatory expert, and commercialization manager, are clearly defined, and the distribution of tasks is balanced. The proposal clearly states some skills that are missing (for example, managing high-volume production, LCA/LCA expertise, and international sales). Plans are in place to hire new people or outsource work in these areas. There is currently a 40% female representation, which is a step toward gender balance. A clear action plan is made to get more women involved in technical and leadership roles during the project. The team is very motivated not only to make the new technology work, but also to be the leader in sustainable cooling technologies in Europe. The long-term business plan, patent strategy, and partnership commitments are all clear signs of this determination and continuity.*

---
*The capacity and role of each team member, the measures in place to ensure motivation and the extent to which the proposed team, as a whole, brings together the necessary knowledge and expertise are very good.*
*The team is led by a multidisciplinary team spanning MOF materials, HVAC productization, industrial scale-up, and blended financing.*
*Gender balance is adequate. Although there is only 1 woman on the board (CEO), a convincing plan was presented to increase the number of women on the board by 2027 (above 50% in executive roles).*
*The proposal consist of convincing missing skills description and recruitment plan in 2026. The connection between the missing skills and the needs of individual Work Packages is insufficiently described. It is minor shortcoming.*
*The description of members of the team is sufficient. They have mapped responsibilities to Work Packages with sufficient way.*
*Which is positive.*

.

---
*The capacity and role of each team member, the measures in place to ensure motivation and the extent to which the proposed team, as a whole, brings together the necessary knowledge and expertise are average.*

*The composition of the team is convincing. It involves a 5 FTE experienced and complementary staff base that spans MOF materials, HVAC/mechanical, controls, field operations, and commercial. It is supported by experienced advisors, including from CNRS.*

*The applicants claim that they commit to ≥50% women in executive roles (CEO/CSO/CTO/COO/CFO) by 2027, ≥33% women on the Board by 2026, and ≥40% women across management by 2028. They also claim that executive searches run 50/50 shortlists with anonymized first-screening and inclusive JD language. However, they do not address gender balance at the organisation level, including non-executive positions. This is a shortcoming.*

*The motivation and capability of the team to implement the innovation proposal and bring it to the market are credible and convincing. This is evidenced by the ecosystem they are nurturing around their innovation supported by an impressive panel of experienced advisors, convincingly substantiated in the proposal.*

*There is a clear plan to acquire additional skills, missing in the company. This is clearly articulated, and involves the recruitment of Pilot-Site Manager, Controls /Software Lead, HVAC-Eng Lead, and a CFO.*

**11. Risk level of the investment (for applicants requesting an investment component):**
**Is the applicant applying for an investment component?**

*Yes*

**Risk level of the investment:**

**If yes, can the company demonstrate that:**
**– The nature and risk level of the investment in the innovation mean that, without an investment from the EIC Fund, European market actors are unwilling to commit the full amount that is needed?**

**AND that:**

**– The company will be able to attract, with the support of the EIC, the remaining funding from other investors within the next two years?**
**Note: if an applicant has previous investors or is in a current investment round, this will not be used as a reason to reject an application against this element. Moreover, this assessment should take into account the international context and whether competitor companies outside of the EU or Associated Countries have access to larger investment amounts.**

---
*The soundness of the arguments proving the risk level of investment is very good.*
*Because it's a deep-tech project that needs a lot of money and takes a long time to pay back, private investors see it as a high-risk investment in the current market. MOF-based advanced material production and the scaling of heat-powered cooling systems are beyond the typical investment horizon of venture capital due to high initial CAPEX, long validation cycles, and certification processes. This makes the contribution of the EIC fund crucial for the investment to materialize. The proposal makes it clear that several industrial partners and investment funds have already sent letters of intent (LoIs), but these investors' contributions depend on the EIC investment being made. The company intends to obtain the remaining funding from private investors and strategic partnerships within two years, subsequent to the mitigation of technological and financial risks with EIC support. The risk profile is clear; the analysis of production scaling, material durability, commercialization time, and cash flow uncertainties is thorough. It has also been shown that EIC support can lower these risks and bring in private capital through a catalytic effect.*

---
*The soundness of the arguments proving the risk level of investment is very good.*

*Through the conditional interest from prospective co-investors and the genuine risk in developing the innovation towards bringing it to market, the company demonstrates the necessity for the EIC investment*

---
*The soundness of the arguments proving the risk level of investment is very good.*

*The applicants are convincing in that they operate in a high-tech sector where the nature and risk level of the investment makes it difficult to attract investors. As such, they are credible in their need for the investment from the EIC Fund, to attract the financial support from European market actors to commit the full amount needed.*

*The company is credible in its ability to attract, with the support of the EIC, the remaining funding from other investors within the next two years. Their current planned investor discussions are clear and convincing. There is a clear indication of the potential investors, with 2 supplying letters of intent. The applicants are overall convincing in their plans to crowd in circa €3 from private co-investors, commercial lenders and national/regional programs.*

**12. Risk mitigation: Have the main risks (e.g. technological, market, financial, regulatory) been identified, together with measures to take to mitigate them?**

---
*The soundness of the risk mitigation strategy is very good.*
*The proposal comprehensively identifies the main risks within the scope of the project and develops appropriate preventive and mitigating strategies for each. Among the technological risks, the long-term cycle endurance of MOF material, production scaling, and potential performance deviations in field integration stand out. To deal with these risks, technical steps like long-term cycle testing, developing alternative formulations, and modular system architecture are planned. For market risks (rate of adoption of new technology, customer reluctance to invest), confidence-building measures such as early field demonstrations, energy performance guarantees, and OEM partnerships are anticipated. Financial risks (increased CAPEX, funding delays) are also modelled to enable the participation of private investors with EIC support, and a local supply chain strategy is developed to reduce costs. For regulatory risks (REACH, SSbD, CE compliance), certification consulting and external verification processes are defined during the project process. Risks are quantitatively assessed in a probability-impact matrix (high-medium-low categories), and both preventive and mitigation measures are separately indicated for each. This holistic approach demonstrates that the project realistically addresses technical and financial uncertainties and progresses at a manageable risk level.*

---
*The soundness of the risk mitigation strategy is very good. There is detailed risk management and mitigation plan included, the risk management includes technological aspect, policy and market details and competition which is positive. The risks has been classified using proper risk taxonomy and scoring. The part concerning implementation risk has been qualified in proper way, is sufficient and well described.*

---
*The soundness of the risk mitigation strategy is very good.*

*The risks have been well described and quantified. The level of details of the identified risks is pertinent. As an example, the applicants have identified a risk linked to value erosion from tariff compression*
*(electricity/water prices drop vs. base case, weakening buyer ROI).*

*The strategy to prevent the risks is convincing and well presented. The applicants have for each identified risk provided convincing risk warning, preventive mitigation, contingency plans and risk owner, and risk triggers.*

*The strategy to mitigate the risks is convincing. The applicants will operate a monthly Risk Committee (CFO/COO) to track Key Risk Indicators (KRIs), with quarterly board reviews. They will also maintain a live dashboard, comprising: tariff indices, investor pipeline, insurer feedback, spec-in count, CE/REACH status, MOF supply KPIs, AND hiring cycle time.*

**13. Implementation plan: Is there a clear implementation plan with defined milestones, work packages and deliverables, together with realistic resources and timings?**

---
*The soundness and credibility of the implementation plan are very good.*
*The project implementation plan gives a mostly well-organized and realistic plan for how to carry out the project, including its stages, work packages, deliverables, and resource allocation. The plan has three main parts: (1) Technological maturation and validation (moving from TRL-5 to TRL-6; laboratory and pilot tests, 12 months), (2) Setting up the FOAK production line and doing field demonstrations (TRL-7/8, 18–24 months), and (3) The commercialization and certification process (TRL-8/9, 12–18 months). Each phase has measurable deliverables, like reaching a certain level of MOF production capacity (kg/month), specific improvement rates in energy efficiency metrics (%ΔPUE, %ΔWUE), and completion dates for certifications. This makes it possible to keep track of progress. The timeline is for 4 to 5 years, which is a good amount of time for deep technology projects, and it makes sense in terms of the order of technical validation, scaling, and entering the market. The resource plan (personnel, budget, subcontracting, equipment investments) is balanced across the work packages, and the partners' responsibilities are clearly spelled out. The FOAK line is also meant to become a commercial production infrastructure after the project is over, which will help ensure that the defined milestones last for a long time. The implementation plan gives a clear picture of how the project will be managed, with clear timelines and resources.*

---
*The soundness and credibility of the implementation plan are very good.*
*Project Management implements quality management procedures and a risk management plan in order to guarantee the successful realization of the project during whole period. There are well defined and addressed several milestones related to the Work Packages. Which is positive. The implementation plan is very detailed and is one of the best part of the proposal. Which is very good.*

---
*The soundness and credibility of the implementation plan are very good.*

*the stages of the implementation plan are convincing and well described. The grant component involves 4 pertinent WPs that span a 24 months duration. The Investment component involves 4 additional WPs that span convincingly the whole duration of the project, i.e., 36 Months.*

*The deliverables are all pertinent and measurable. The project involves 9 deliverables, 3 of which with a public dissemination reach.*

*The time scale is realistic and credible. The project spans an appropriate 36 months duration.*

*The resources needed to deliver the grant agreement are well defined and justified.*

*The defined milestones are pertinent and sustainable in the long-run. Six (6) milestones relate to the grant element, while 4 additional milestones are dedicated to the Investment component. All milestones are credible and convincing.*

## Recommended for interview:

Score: **0.00** (Threshold: 9 / 9.00 , Weight: - )

## Scope of the application

**Based on the information provided, this application is:**
*'in scope' because it corresponds, wholly or in part, to the topic description against which it has been submitted*

**Comments (in case the proposal is out of scope)**
*Not provided*

## Use of human embryonic stem cells (hESC)

**Does this proposal involve the use of hESC?**
*No*

**If YES, please state whether the use of hESC is, or is not, in your opinion, necessary to achieve the scientific objectives of the proposal and the reasons why. Alternatively, please state if it cannot be assessed whether the use of hESC is necessary or not, because of a lack of information.**
*Not provided*

## Use of human embryos

**Does this proposal involve the use of human embryos?**
*No*

**If YES, please explain how the human embryos will be used in the project.**
*Not provided*

## Activities excluded from funding

**Activities that :**
**- aim at human cloning for reproductive purposes, or**
**- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or**
**- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer, or**
**- lead to the destruction of human embryos (for example, for obtaining stem cells)**

**are excluded from funding. Does the proposal include any of these activities?**

*No*

**If YES, please explain.**

*Not provided*

## Do no significant harm principle

Status: **Yes**

**If Partially/No/Cannot be assessed please explain**

*Not provided*

## Exclusive focus on civil applications

**Do the activities proposed have an exclusive focus on civil applications (activities intended to be used in military application or aims to serve military purposes cannot be funded)?**

*Yes*

**If NO, please explain.**

*Not provided*

## Artificial Intelligence

**Do the activities proposed involve the use and/or development of AI-based systems and/or techniques?**

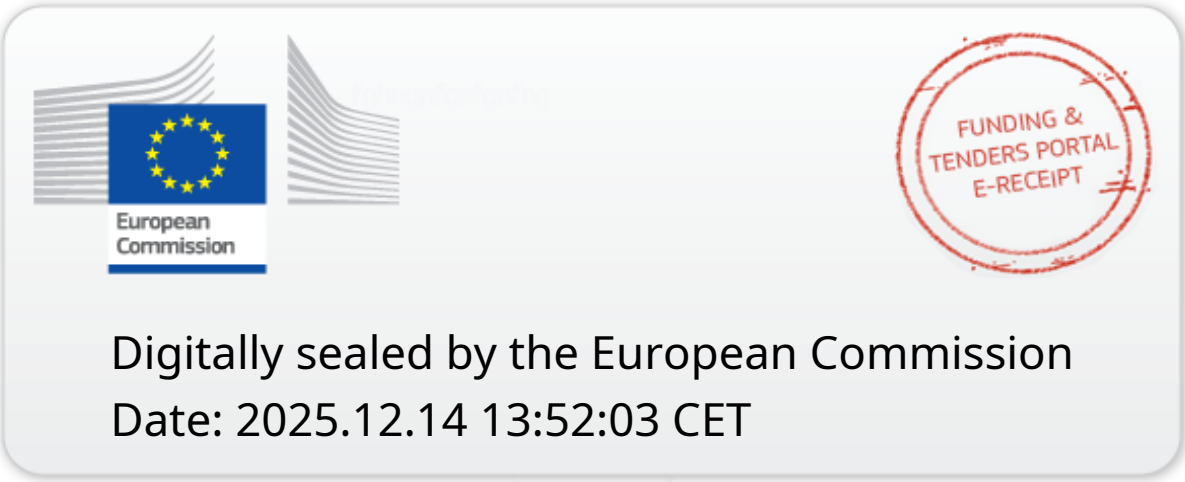
*No*

**If YES, the technical robustness of the proposed system must be evaluated under the appropriate criterion.**

*Not provided*

## Consensus meeting comments (if applicable)

*The proposal did not reach the conditions necessary to be considered for the consensus meeting as per the EIC Work Programme 2025 rules - "If two of the three evaluators give a GO for all the evaluation elements, then there will be a consensus meeting to decide if you will be invited to an interview with an EIC Jury".*




This electronic receipt is a digitally signed version of the document submitted by your organisation. Both the content of the document and a set of metadata have been digitally sealed.

This digital signature mechanism, using a public-private key pair mechanism, uniquely binds this eReceipt to the modules of the Funding & Tenders Portal of the European Commission, to the transaction for which it was generated and ensures its full integrity. Therefore a complete digitally signed trail of the transaction is available both for your organisation and for the issuer of the eReceipt.

Any attempt to modify the content will lead to a break of the integrity of the electronic signature, which can be verified at any time by clicking on the eReceipt validation symbol.

More info about eReceipts can be found in the FAQ page of the Funding & Tenders Portal.

(https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/support/faq)